ST MARY'S DSG, KLOOF

GRADE: 10 **DATE**: 6 February 2019

MATHEMATICS – No Systems, Patterns, Algebra

TIME: 40 min **TOTAL**: 53

Question 1

a) For which values of x where $x \in \{-2\,;0\,;2\,;4\}$ is $\sqrt{\frac{9}{2-x}}$ (4)

1) A irrational number? 0

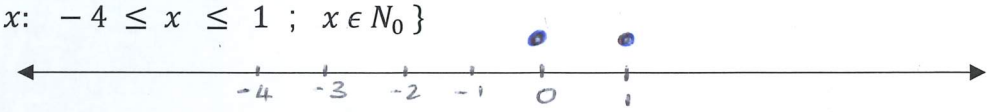
2) A non real number? 4

3) A rational number ? -2

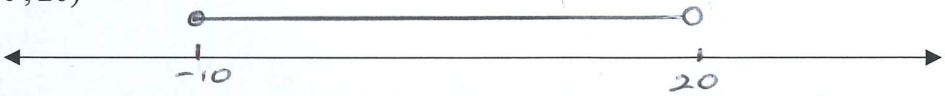
4) Undefined 2

b) Represent the following on a number line:

1) $\{x:\ -4 \leq x \leq 1\ ;\ x \in N_0\}$ (2)



2) $[-10\,;20)$ (2)



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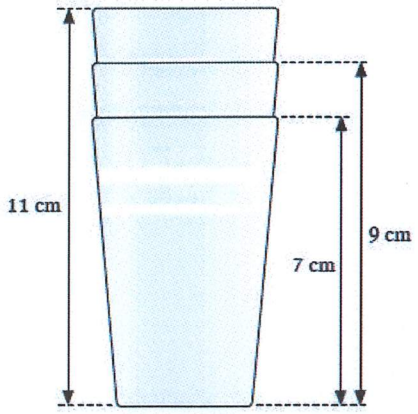
Question 2

a) Fill in the missing spaces in the following number patterns:

1) -1 ; 2 ; 5 ; 8 ; 11 ; 14 (2)

2) 2 ; 3 ; 5 ; 8 ; 12 ; 17 ; 23 (2)

b) You are stacking polystyrene cups, which fit into each other as in the diagram below. The 1st cup is 7 cm high. Two cups stacked together are 9 cm , and three cups stacked together are 11 cm high.



1) How high would 4 cups be ? (1)

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2) Find the general formula for the height of a stack of n cups. (2)

$T_n = 2n + 5$

3) How many cups will stack on a shelf if the height between the shelves is 21 cm ? (2)

$2n + 5 = 21$

$n = 8$

4) A box is 30 cm high and can only fit a single stack of cups. You need 100 cups for a party, how many boxes would you need to buy? (3)

no of cups = 12 ✓

$\frac{100}{12} = 8,3$ ✓ 9 boxes ✓

c) **4 ; 7 ; 14 ; 25.....** 40 59

1) Determine the next 2 terms (2)

3 7 11 15 19 40 ; 59.

4 4 4 4

2) Determine the general term in the form of $T_n = an^2 + bn + c$ (4)

$2a = 4$ $\quad 3(2) + b = 3$ $\quad -1 + c = 4$

$a = 2$ $\quad b = -3$ $\quad c = 5$

$T_n = 2n^2 - 3n + 5.$

d) Consider the following number Pattern: $\mathbf{0,\dot{1}}$; $\mathbf{0,\dot{2}}$; $\mathbf{0,\dot{3}}$ ……

1) Determine the next two terms $0,4\checkmark$; $0,5\checkmark$ (2)

2) Write down T_3 in the form $\frac{a}{b}$ $3/9\checkmark = 1/3.$ (3)

3) Write down 2 rational numbers between the 2nd and 3rd term. (2)

$0,21$; $0,22$; $0,23.$

4) Determine the largest number in the sequence where $T_n < 1$. (1)

$0,\dot{9}$

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Question 3

a) Multiply and simplify

1) $5\left(\frac{1}{5}x - 1\right)\left(5x - \frac{1}{5}\right)$ (3)

$5\left(x^2 - \frac{1}{25}x - 5x + \frac{1}{5}\right)$

$= 5x^2 - \frac{1}{5}x - 25x + 1 = 5x^2 - \frac{126}{5}x + 1$

2) $-(2x - 3)(4x^2 + 6x + 9)$ (3)

$= -(8x^3 - 27)$

$= -8x^3 + 27$

3) $(3^x + 1)^2$ (2)

$(3^x)^2 + 2 \cdot 3^x + 1$

$3^{2x} + 2 \cdot 3^x + 1$ or $9^x + 2 \cdot 3^x + 1$

b) Factorise fully

1) $3a^2 - 12\,b^2$ $3(a+2b)(a-2b)$ (2)

2) $(a - 2b)^3 - 4\,(a - 2\,b)$ (3)

$(a-2b)[(a-2b)^2 - 4]$

$(a-2b)\ [(a-2b+2)(a-2b-2)]$

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Question 4

a) If **0 ; 7 ; 26 ; 63 ; 124.....**
Determine the general term of the number pattern. (2)

$T_n = n^3 - 1$

b) A magic substance doubles it volume every minute. A small quantity is placed in a bottle and at 17h13 is exactly full. At what time is the container a quarter full (2)

17h11

c) Find the product of : (2)

$\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{5}\right)\left(1-\frac{1}{6}\right)\left(1-\frac{1}{7}\right)\ldots\ldots\ldots\ldots\ldots\ldots\left(1-\frac{1}{74}\right)\left(1-\frac{1}{75}\right)$

$\left(\frac{2}{3}\right)\left(\frac{3}{4}\right)\left(\frac{4}{5}\right)$ . . . . . $\left(\frac{73}{74}\right)\left(\frac{74}{75}\right)$

$\frac{2}{75}$

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